

Shi Chen

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EDUCATION

ETH Zurich

Sep. 2022 – Present

M.Sc. in Electrical Engineering and Information Technology | **GPA: 5.63/6.0**

Zurich, Switzerland

- **Courses:** 3D Vision, Image Analysis and Computer Vision, Mixed Reality, Vision Algorithms for Mobile Robotics, Robot Learning, Computer Vision and Artificial Intelligence for Autonomous Cars, etc.

Kyoto University

Apr. 2018 – Mar. 2022

B.Eng. in Electrical and Electronic Engineering | **GPA: 3.94/4.3**

Kyoto, Japan

- **Courses:** Machine Learning, Pattern Recognition, Computer Software, etc.

RESEARCH EXPERIENCE

MonoDy-GS: Online Monocular Dynamic Gaussian Splatting

Apr. 2024 – Aug. 2024

Advised by Prof. Martin Oswald, Dr. Sandro Lombardi and Prof. Marc Pollefeys

ETH Zurich

- Developed online pipeline capable of modeling dynamic scenes from RGB-D sequence using 3D Gaussians;
- Incorporated various priors based on real-world scene dynamics to address lack of multi-view coverage.

EvenNICER-SLAM: Event-based Neural Implicit Encoding SLAM

Oct. 2022 – Jan. 2023

Advised by Dr. Danda Pani Paudel and Prof. Luc Van Gool

ETH Zurich

- Enhanced robustness of NICE-SLAM against fast camera motion by integration of event data input;
- EvenNICER-SLAM significantly outperforms NICE-SLAM in tracking with reduced RGB-D input frequency.

Road Scene Pedestrian Relocation for Data Augmentation

May 2021 – Feb. 2022

Advised by Prof. Shohei Nobuhara and Prof. Ko Nishino

Kyoto University

- Devised data augmentation method that automatically cuts out large-scale pedestrians in foreground of road-scene videos and relocates them at farther positions with correct scale and occlusion;
- Proposed method improves Mask R-CNN in both detection and instance segmentation of far-away pedestrians.

COURSE PROJECTS

NeRaser: NeRF-based 3D Object Eraser | *Mixed Reality*

Oct. 2023 – Jan. 2024

- Developed object removal pipeline in NeRF-based scene representation trained on casual monocular videos;
- NeRaser ensures visual consistency of occluded area through 3D-aware inpainting strategy.

Monocular Visual Odometry | *Vision Algorithms for Mobile Robotics*

Nov. 2023 – Jan. 2024

- Implemented monocular feature-matching-based visual odometry pipeline in Matlab.

Hierarchical Dense Neural Point Cloud-based SLAM | *3D Vision*

Mar. 2023 – Jun. 2023

- Enhanced robustness of Point-SLAM through coarse-to-fine optimization using multiple levels of neural points;
- Achieved significant improvement in tracking on real-world dataset ScanNet.

Generalizable 3D Reconstruction Using Correspondence Matching | *Robot Learning*

Mar. 2023 – Aug. 2023

- Proposed generalizable SDF-based sparse-view 3D reconstruction method utilizing 2D feature matching.

COMPETITIONS & AWARDS

Ichikawa International Scholarship

Oct. 2020 – Sep. 2021

Champion, Summer Camp Robot Competition 2019 | *Kyoto University*

Sep. 2019

Champion, Summer Camp Robot Competition 2018 | *Kyoto University*

Sep. 2018

Monbukagakusho Honors Scholarship

Apr. 2018 – Mar. 2019

SKILLS

Programming: Python, Matlab, PyTorch, Git, Docker, Singularity, Arduino

Languages: Chinese (native), Japanese (native), English (Fluent)